

DS Transition Roadman

Cesar Aguirre | Start: April 21, 2026 | Target: July 14, 2026 | Goal: Tier 1 Finance DS Roles

Master Timeline

Phase	Dates	Deliverable
1 — Foundation	Apr 21 – May 3	Environment set up, SQL practice started, core ML concepts reviewed
2 — Project 1	May 4 – May 24	Credit Default Model (Give Me Some Credit) live on GitHub
3 — Project 2	May 25 – Jun 7	Fraud Detection Model (Kaggle CC Fraud) live on GitHub
4 — Project 3	Jun 8 – Jun 14	Credit Default API deployed live via FastAPI + Render
5 — AWS	Jun 15 – Jun 28	AWS Cloud Practitioner cert earned; SageMaker hands-on complete
6 — Interview + Apply	Jun 29 – Jul 13	Applications submitted to Tier 1 companies; interview-ready

Phase 1 — Foundation

Week 1 — Apr 21–26

- Set up Python environment: Anaconda, virtual environment, VS Code or Jupyter Lab
- Create professional GitHub profile — photo, bio mentioning finance + data science
- Review pandas and scikit-learn: loading data, cleaning nulls, encoding, train/test split
- Resource: *Python for Data Analysis* by Wes McKinney — Chapters 1–7 only
- Sign up for StrataScratch (free tier) — complete 5 finance-tagged SQL problems at medium difficulty
- Time commitment: ~1.5 hours/day

Week 2 — Apr 28–May 3

- Review core ML concepts: logistic regression, decision trees, XGBoost, ROC/AUC, confusion matrix, precision vs. recall
- Write a plain-English paragraph explaining each concept — builds your interview answer muscle
- SQL practice continues: 5 more StrataScratch problems, focus on window functions and CTEs
- Set up clean GitHub project folder structure — signals professionalism to hiring managers
- Time commitment: ~1.5 hours/day

StrataScratch: Start with the free tier. Only upgrade if you find yourself consistently stuck with no way to debug why.

Phase 2 — Project 1: Credit Default Model

Dataset: Give Me Some Credit (Kaggle) | Libraries: pandas, scikit-learn, XGBoost, matplotlib/seaborn

Week 3 — May 4–10 | Data Exploration

- Download Give Me Some Credit from Kaggle
- Exploratory data analysis: distributions, missing values, correlation matrix, default rate (~6.7%)
- Handle missing values — two columns have ~20% nulls; justify your imputation strategy
- Commit EDA notebook to GitHub with a clear README explaining the dataset and prediction goal
- Time commitment: ~1.5 hours/day

Week 4 — May 11–17 | Modeling

- Build three models: logistic regression (baseline), random forest, XGBoost
- Handle class imbalance using SMOTE or class_weight — document why this matters
- Evaluate with ROC/AUC, precision-recall curve, and confusion matrix
- Add feature importance chart — connects directly to your Citi credit analysis experience
- Time commitment: ~2 hours/day

Week 5 — May 18–24 | Polish and Write-Up

- Clean notebook: remove dead cells, add markdown commentary explaining each decision
- Write professional README: problem statement, dataset, methodology, results, how to run
- Add a section explicitly connecting findings to real-world credit risk — your key differentiator
- Push final version to GitHub
- Time commitment: ~1 hour/day

Phase 3 — Project 2: Fraud Detection Model

Dataset: Kaggle Credit Card Fraud Detection | Key focus: class imbalance (0.17% fraud), precision-recall tradeoff

Week 6 — May 25–31 | Exploration + Modeling

- Download Kaggle Credit Card Fraud Detection dataset
- EDA: visualize extreme class imbalance, understand PCA-transformed features
- Build logistic regression baseline, then XGBoost with SMOTE
- Document precision vs. recall tradeoff — missing real fraud (false negative) costs more than a false positive
- Time commitment: ~1.5 hours/day

Week 7 — Jun 1–7 | Threshold Tuning + Polish

- Experiment with classification threshold — show moving from 0.5 to 0.3 catches more fraud at cost of more false positives
- Connect README explicitly to your Citi fraud review experience — this is your unique angle
- Clean, finalize, push to GitHub
- Time commitment: ~1.5 hours/day

Interview talking point: 'In my independent assessment role I reviewed credit card fraud cases — this project let me model the detection problem I have evaluated operationally.'

Phase 4 — Project 3: Model Deployment

Week 8 — Jun 8–14

- Take your credit default model and deploy it as a FastAPI REST API
- Step 1: Save model with joblib
- Step 2: Build FastAPI app — accepts borrower inputs, returns default probability score
- Step 3: Deploy to Render.com (free tier) — they give you a live public URL
- Add README with live URL and screenshot of a sample API response

- Time commitment: ~2 hours/day first 3 days, lighter the rest of the week

Resume line after this week: 'Developed and deployed a credit default prediction model as a REST API using FastAPI and scikit-learn, hosted on Render.' This is when your resume gains the word deployed.

Phase 5 — AWS + SageMaker

Week 9 — Jun 15–21 | Cloud Practitioner Prep

- Study resource: AWS Skill Builder (free) + Stephane Maarek's Cloud Practitioner course on Udemy (~\$15 on sale)
- Focus areas: core services (S3, EC2, Lambda), cloud concepts, security basics, pricing model
- Practice exams: Tutorials Dojo — best available practice tests (~\$15)
- This exam is conceptual, not coding — genuinely approachable with 1–2 weeks of prep
- Time commitment: ~1.5 hours/day

Week 10 — Jun 22–28 | SageMaker Hands-On + Exam

- Schedule and take AWS Cloud Practitioner exam (Pearson VUE online proctored — from home)
- Spend 2–3 sessions running your credit default model through a SageMaker notebook on the free tier
- Goal: be able to describe the SageMaker experience accurately in an interview
- Add AWS Cloud Practitioner to resume and LinkedIn upon passing
- Time commitment: ~1.5 hours/day

Phase 6 — Interview Prep + Applications

Week 11 — Jun 29–Jul 5 | Technical Interview Prep

- SQL: final push on StrataScratch — aim for 30 total finance problems completed across the roadmap
- Practice explaining all three projects out loud — record yourself once and watch it back
- Study Capital One / SoFi specific topics: champion/challenger testing, scorecard development, Gini coefficient in credit models, vintage analysis
- Resource: *Credit Risk Scorecards* by Naeem Siddiqi — Chapters 1–4 only for interview vocabulary
- Time commitment: ~1.5 hours/day

Week 12 — Jul 6–13 | Resume Finalization + Applications

- Update resume: add all three projects, AWS cert, TypeScript and deployment language
- Tailor summary for each company — SoFi version emphasizes consumer lending lifecycle; Capital One version emphasizes fraud and ML modeling
- Apply to Tier 1 positions: SoFi, Capital One, Discover, Upstart, LendingClub, American Express
- Reach out to 2–3 DS professionals at each target company on LinkedIn for genuine conversations — referrals at Capital One move resumes from general pile to recruiter's desk
- Time commitment: ~1.5 hours/day

Tier 1 Target Companies

Company	Role Type	Fit Strength
SoFi	Credit Strategy DS	Strongest — consumer lending lifecycle match
Capital One	Principal Associate DS	Strong — fraud + ML modeling angle

Discover	Risk Data Scientist	Strong — credit card + compliance background
Upstart	Credit / Risk DS	Strong — ML-first lending company
LendingClub	Credit Risk DS	Strong — consumer lending focus
American Express	Risk Analytics DS	Strong — large-scale card portfolio

Mock Interview

When you reach Week 11, return for a full mock technical interview simulating Capital One's or SoFi's actual format. By that point you will have real projects to discuss, which makes the practice far more valuable than doing it earlier. Questions will cover SQL, ML concepts, your project decisions, and credit strategy scenarios.

Prepared April 21, 2026 | Tier 1 Target Ready-to-Apply Date: July 14, 2026